

Problem A. One-stroke Path

Time Limit 2000 ms

Problem Statement

You are given an undirected unweighted graph with N vertices and M edges that contains neither self-loops nor double edges.

Here, a *self-loop* is an edge where $a_i = b_i$ ($1 \leq i \leq M$), and *double edges* are two edges where $(a_i, b_i) = (a_j, b_j)$ or $(a_i, b_i) = (b_j, a_j)$ ($1 \leq i < j \leq M$).

How many different paths start from vertex 1 and visit all the vertices exactly once?

Here, the endpoints of a path are considered visited.

For example, let us assume that the following undirected graph shown in Figure 1 is given.

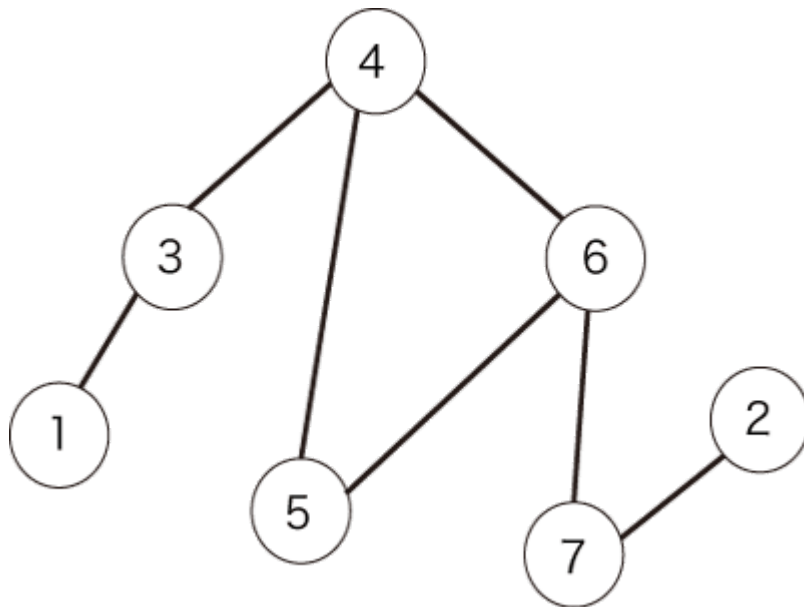


Figure 1: an example of an undirected graph

The following path shown in Figure 2 satisfies the condition.

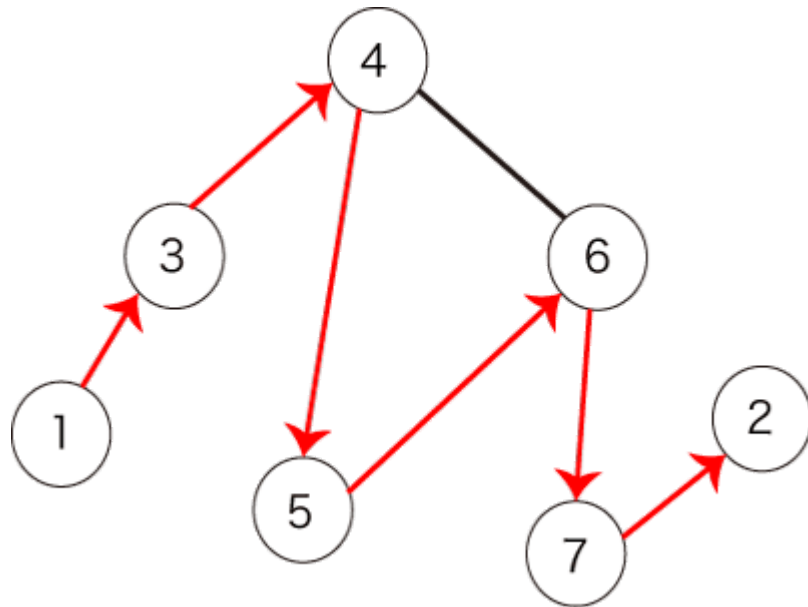


Figure 2: an example of a path that satisfies the condition

However, the following path shown in Figure 3 does not satisfy the condition, because it does not visit all the vertices.

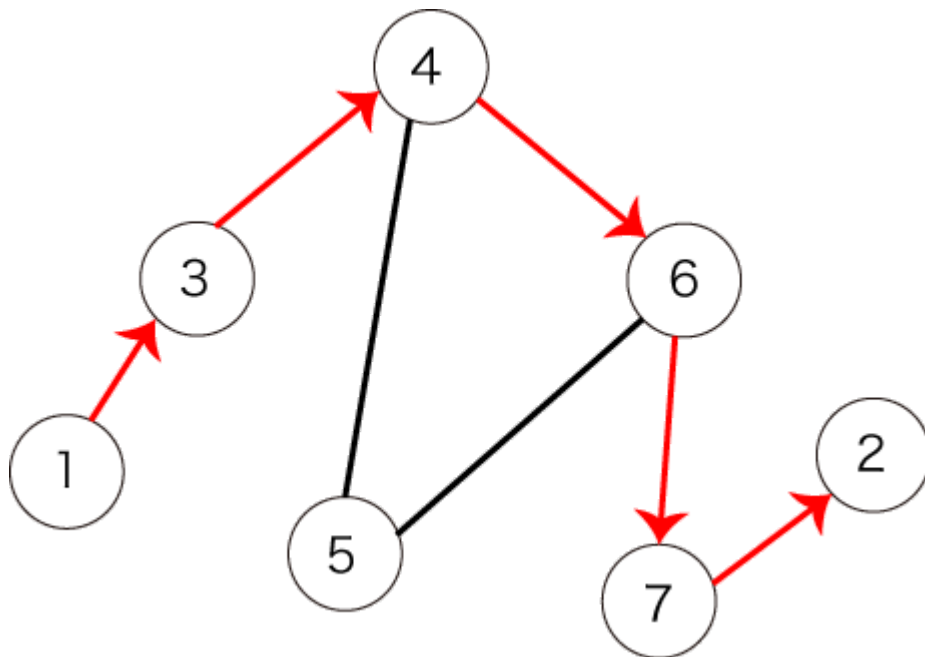


Figure 3: an example of a path that does not satisfy the condition

Neither the following path shown in Figure 4, because it does not start from vertex 1.

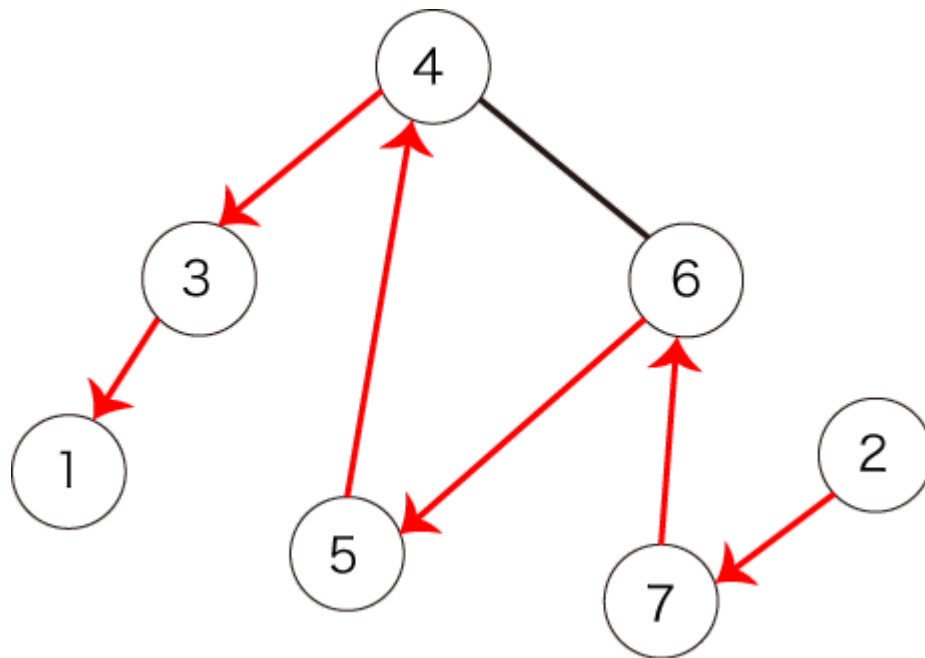


Figure 4: another example of a path that does not satisfy the condition

Constraints

- $2 \leq N \leq 8$
- $0 \leq M \leq N(N - 1)/2$
- $1 \leq a_i < b_i \leq N$
- The given graph contains neither self-loops nor double edges.

Input

The input is given from Standard Input in the following format:

```

N M
a1 b1
a2 b2
:
aM bM

```

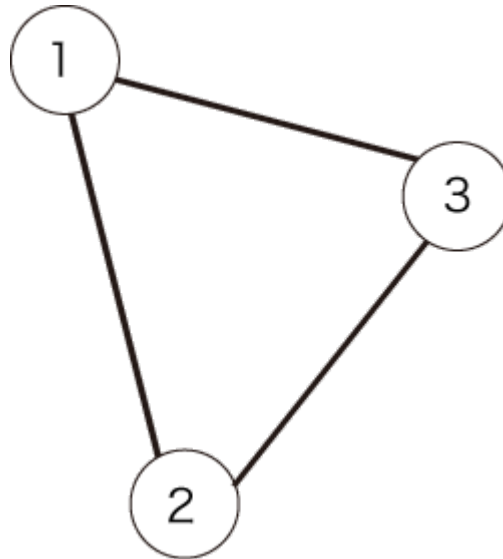
Output

Print the number of the different paths that start from vertex 1 and visit all the vertices exactly once.

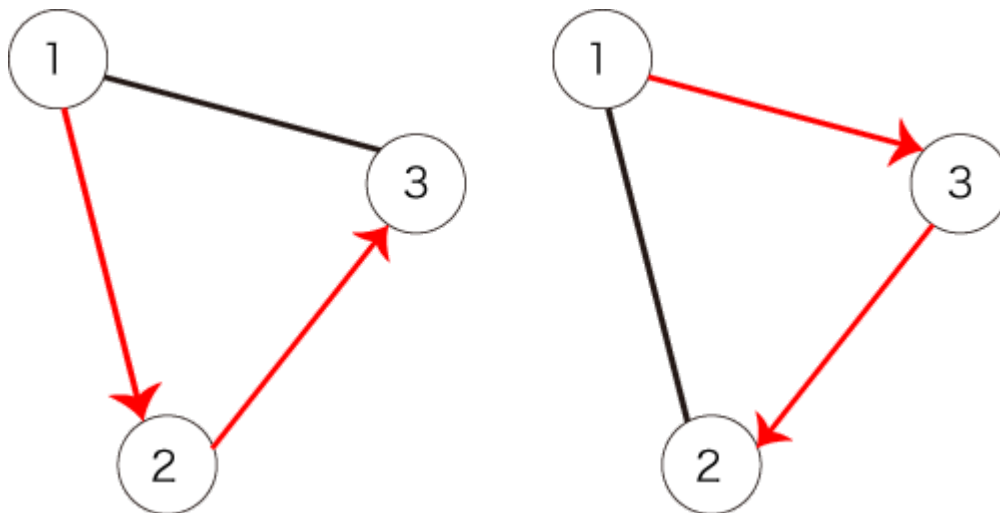
Sample 1

Input	Output
3 3 1 2 1 3 2 3	2

The given graph is shown in the following figure:



The following two paths satisfy the condition:

**Sample 2**

Input	Output
7 7 1 3 2 7 3 4 4 5 4 6 5 6 6 7	1

This test case is the same as the one described in the problem statement.

Problem B. Ladder Takahashi

Time Limit 2000 ms

Problem Statement

There is a 10^9 -story building with N ladders.

Takahashi, who is on the 1-st (lowest) floor, wants to reach the highest floor possible by using ladders (possibly none).

The ladders are numbered from 1 to N , and ladder i connects the A_i -th and B_i -th floors. One can use ladder i in either direction to move from the A_i -th floor to the B_i -th floor or vice versa, but not between other floors.

Takahashi can freely move within the same floor, but cannot move between floors without using ladders.

What is the highest floor Takahashi can reach?

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq A_i, B_i \leq 10^9$
- $A_i \neq B_i$
- All values in the input are integers.

Input

The input is given from Standard Input in the following format:

```
N
A1 B1
A2 B2
...
AN BN
```

Output

Print an integer representing the answer.

Sample 1

Input	Output
4 1 4 4 3 4 10 8 3	10

He can reach the 10-th floor by using ladder 1 to get to the 4-th floor and then ladder 3 to get to the 10-th floor.

Sample 2

Input	Output
6 1 3 1 5 1 12 3 5 3 12 5 12	12

Sample 3

Input	Output
3 500000000 600000000 600000000 700000000 700000000 800000000	1

He may be unable to move between floors.

Problem C. Relative Position

Time Limit 2500 ms

Problem Statement

There are N people numbered 1 to N on a coordinate plane.

Person 1 is at the origin.

You are given M pieces of information in the following form:

- From person A_i 's perspective, person B_i is X_i units away in the positive x -direction and Y_i units away in the positive y -direction.

Determine the coordinates of each person. If the coordinates of a person cannot be uniquely determined, report that fact.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $0 \leq M \leq 2 \times 10^5$
- $1 \leq A_i, B_i \leq N$
- $A_i \neq B_i$
- $-10^9 \leq X_i, Y_i \leq 10^9$
- All input values are integers.
- The given information is consistent.

Input

The input is given from Standard Input in the following format:

```
N M
A1 B1 X1 Y1
⋮
AM BM XM YM
```

Output

Print N lines.

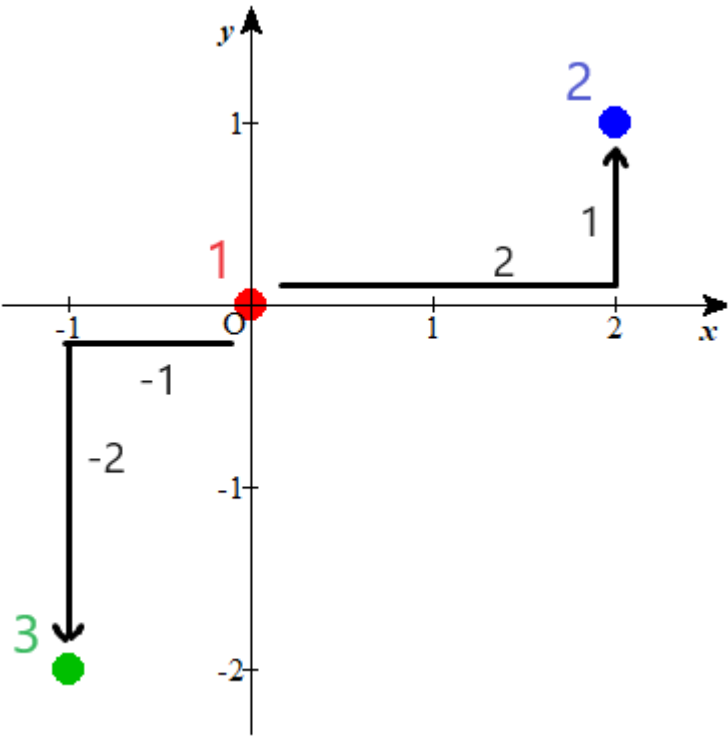
If the coordinates of person i cannot be uniquely determined, the i -th line should contain `undecidable`.

If they can be uniquely determined as (s_i, t_i) , the i -th line should contain s_i and t_i in this order, separated by a space.

Sample 1

Input	Output
3 2 1 2 2 1 1 3 -1 -2	0 0 2 1 -1 -2

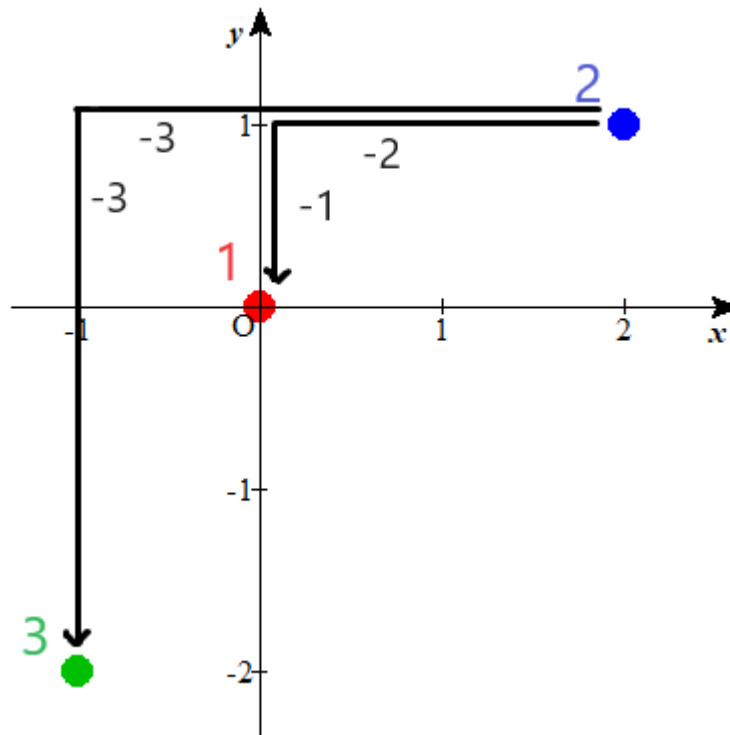
The figure below shows the positional relationship of the three people.



Sample 2

Input	Output
3 2 2 1 -2 -1 2 3 -3 -3	0 0 2 1 -1 -2

The figure below shows the positional relationship of the three people.



Sample 3

Input	Output
5 7 1 2 0 0 1 2 0 0 2 3 0 0 3 1 0 0 2 1 0 0 3 2 0 0 4 5 0 0	0 0 0 0 0 0 undecidable undecidable

The same piece of information may be given multiple times, and multiple people may be at the same coordinates.

Problem D. Bridge

Time Limit 2000 ms

Problem Statement

You are given an undirected connected graph with N vertices and M edges that does not contain self-loops and double edges.

The i -th edge ($1 \leq i \leq M$) connects Vertex a_i and Vertex b_i .

An edge whose removal disconnects the graph is called a *bridge*.

Find the number of the edges that are bridges among the M edges.

Notes

- A *self-loop* is an edge i such that $a_i = b_i$ ($1 \leq i \leq M$).
- *Double edges* are a pair of edges i, j such that $a_i = a_j$ and $b_i = b_j$ ($1 \leq i < j \leq M$).
- An undirected graph is said to be *connected* when there exists a path between every pair of vertices.

Constraints

- $2 \leq N \leq 50$
- $N - 1 \leq M \leq \min(N(N - 1)/2, 50)$
- $1 \leq a_i < b_i \leq N$
- The given graph does not contain self-loops and double edges.
- The given graph is connected.

Input

Input is given from Standard Input in the following format:

```
N M
a1 b1
a2 b2
:
aM bM
```

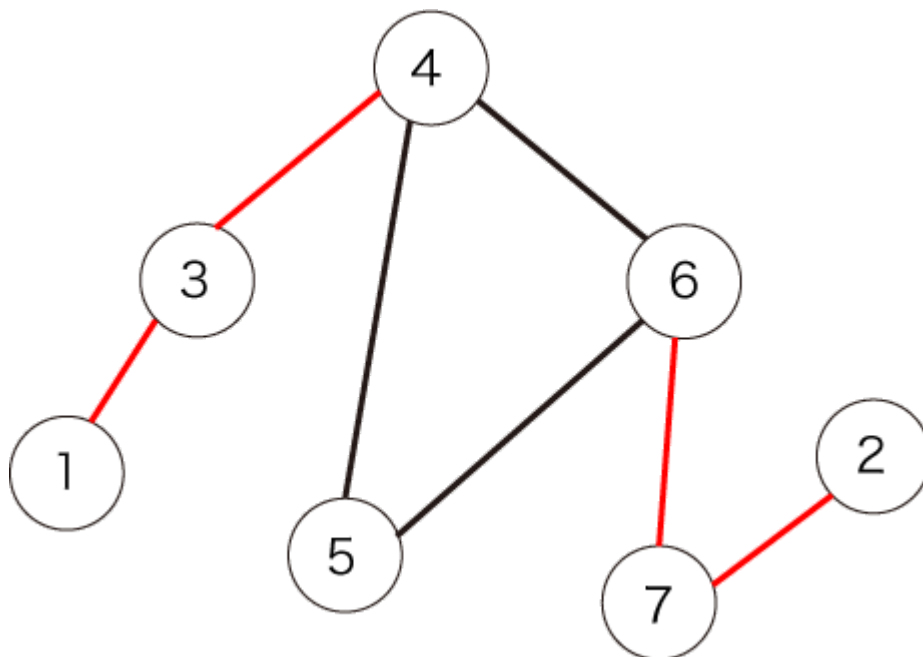
Output

Print the number of the edges that are bridges among the M edges.

Sample 1

Input	Output
<pre> 7 7 1 3 2 7 3 4 4 5 4 6 5 6 6 7 </pre>	4

The figure below shows the given graph:



The edges shown in red are bridges. There are four of them.

Sample 2

Input	Output
<pre> 3 3 1 2 1 3 2 3 </pre>	0

It is possible that there is no bridge.

Sample 3

Input	Output
6 5 1 2 2 3 3 4 4 5 5 6	5

It is possible that every edge is a bridge.

Problem E. Takahashi's Secret

Time Limit 2000 ms

Problem Statement

Takahashi has N friends. They have nicknames: Friend 1, Friend 2, $\dots$, Friend N .

One day, Takahashi accidentally let one of his friends, Friend X , learn his shameful secret. For each $i = 1, 2, \dots, N$, when Friend i learns the secret, he/she will share it with Friend A_i , if Friend A_i has not already learned it.

How many of Takahashi's friends will learn the secret in the end?

Constraints

- $2 \leq N \leq 10^5$
- $1 \leq X \leq N$
- $1 \leq A_i \leq N$
- $A_i \neq i$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
N X
A_1 A_2  $\dots$  A_N
```

Output

Print the answer.

Sample 1

Input	Output
4 2 3 1 1 2	3

Takahashi's secret will be learned by Friend 1, Friend 2, and Friend 3, as follows.

- One day, Takahashi let Friend 2 learn the secret.
- Friend 2 shares it with Friend 1.
- Friend 1 shares it with Friend 3.

In the end, three of his friends learn the secret, so we print 3.

Sample 2

Input	Output
20 12 7 11 10 1 7 20 14 2 17 3 2 5 19 20 8 14 18 2 10 10	7

Problem F. Remembering the Days

Time Limit 2000 ms

Problem Statement

A region has N towns numbered 1 to N , and M roads numbered 1 to M .

The i -th road connects town A_i and town B_i bidirectionally with length C_i .

Find the maximum possible total length of the roads you traverse when starting from a town of your choice and getting to another town without passing through the same town more than once.

Constraints

- $2 \leq N \leq 10$
- $1 \leq M \leq \frac{N(N-1)}{2}$
- $1 \leq A_i < B_i \leq N$
- The pairs (A_i, B_i) are distinct.
- $1 \leq C_i \leq 10^8$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N M
A1 B1 C1
⋮
AM BM CM
```

Output

Print the answer.

Sample 1

Input	Output
4 4 1 2 1 2 3 10 1 3 100 1 4 1000	1110

If you travel as $4 \rightarrow 1 \rightarrow 3 \rightarrow 2$, the total length of the roads you traverse is 1110.

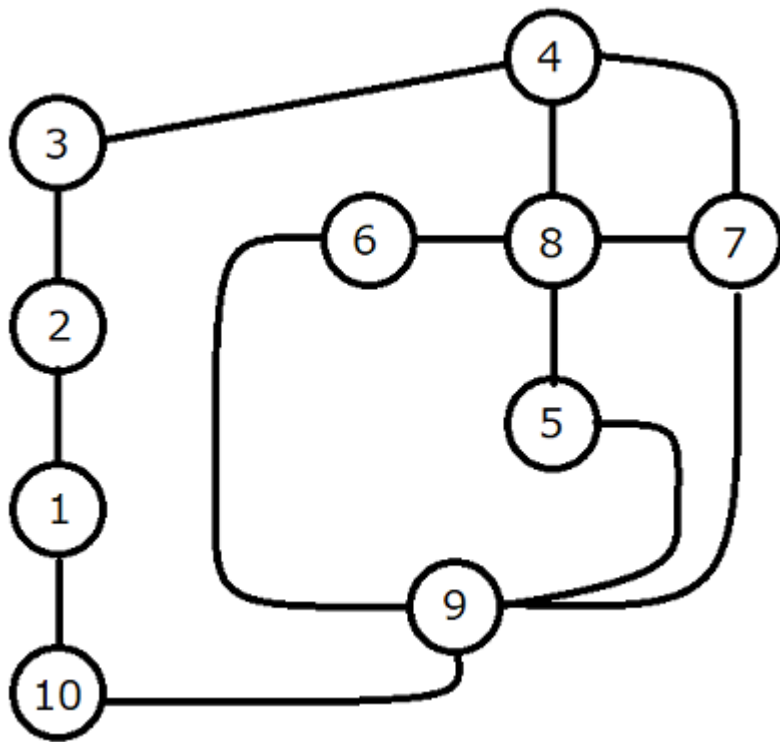
Sample 2

Input	Output
10 1 5 9 1	1

There may be a town that is not connected to a road.

Sample 3

Input	Output
10 13 1 2 1 1 10 1 2 3 1 3 4 4 4 7 2 4 8 1 5 8 1 5 9 3 6 8 1 6 9 5 7 8 1 7 9 4 9 10 3	20



Problem G. A..B..C

Time Limit 2000 ms

Problem Statement

A string S is given.

Find how many places in S have **A**, **B**, and **C** in this order at even intervals.

Specifically, find the number of triples of integers (i, j, k) that satisfy all of the following conditions. Here, $|S|$ denotes the length of S , and S_x denotes the x -th character of S .

- $1 \leq i < j < k \leq |S|$
- $j - i = k - j$
- $S_i = \text{A}$
- $S_j = \text{B}$
- $S_k = \text{C}$

Constraints

- S is an uppercase English string with length between 3 and 100, inclusive.

Input

The input is given from Standard Input in the following format:

S

Output

Print the answer.

Sample 1

Input	Output
AABCC	2

There are two triples $(i, j, k) = (1, 3, 5)$ and $(2, 3, 4)$ that satisfy the conditions.

Sample 2

Input	Output
ARC	0

Sample 3

Input	Output
AABAAABBAEDCCCD	4

Problem H. Make it Simple

Time Limit 2000 ms

Problem Statement

You are given an undirected graph with N vertices and M edges, where the vertices are numbered 1 through N and the edges are numbered 1 through M . Edge i connects vertices u_i and v_i .

To make the graph simple by removing edges, what is the minimum number of edges that must be removed?

Here, a graph is called simple if and only if it does not contain self-loops or multi-edges.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $0 \leq M \leq 5 \times 10^5$
- $1 \leq u_i \leq N$
- $1 \leq v_i \leq N$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
 $N$   $M$   
 $u_1$   $v_1$   
 $u_2$   $v_2$   
 $\vdots$   
 $u_M$   $v_M$ 
```

Output

Print the minimum number of edges that must be removed to make the graph simple.

Sample 1

Input	Output
3 5 1 2 2 3 3 2 3 1 1 1	2

By removing edges 3 and 5, the graph becomes simple. This is one of the ways to remove the minimum number of edges, so the answer is 2.

Sample 2

Input	Output
1 0	0

Sample 3

Input	Output
6 10 6 2 4 1 5 1 6 6 5 3 5 1 1 4 6 4 4 2 5 6	3

Problem I. Takahashi Tour

Time Limit 2000 ms

Problem Statement

The Republic of AtCoder has N cities numbered 1 through N and $N - 1$ roads numbered 1 through $N - 1$. Road i connects City A_i and City B_i bidirectionally. It is guaranteed that one can travel between every pair of cities using roads.

Takahashi will depart City 1 and have a journey by repeating the following.

- If there are unvisited cities that are directly connected to the city Takahashi is in now, he goes to the city with the smallest number among those cities.
- Otherwise,
 - if he is in City 1, he ends the journey;
 - otherwise, he goes to the city he was in just before visiting the current city for the first time.

Find the sequence of cities visited by Takahashi in the order he visits them.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq A_i, B_i \leq N$
- It is possible to travel between every pair of cities using roads.

Input

Input is given from Standard Input in the following format:

```
N
A1 B1
⋮
AN-1 BN-1
```

Output

Print the sequence of cities visited by Takahashi in the order he visits them, including City 1 at the beginning and end of the journey, with spaces in between.

Sample 1

Input	Output
4 1 2 4 2 3 1	1 2 4 2 1 3 1

His journey will be as follows.

- Start in City 1.
- The unvisited cities directly connected to City 1 are City 2 and 3. Go to the city with the smaller number, that is, City 2.
- The unvisited city directly connected to City 2 is City 4. Go there.
- There is no unvisited city directly connected to City 4. Go back to City 2.
- There is no unvisited city directly connected to City 2. Go to City 1, where he was in just before visiting City 2 for the first time.
- The unvisited city directly connected to City 1 is City 3. Go there.
- There is no unvisited city directly connected to City 3. Go back to City 1.
- There is no unvisited city directly connected to City 1. End the journey.

Sample 2

Input	Output
5 1 2 1 3 1 4 1 5	1 2 1 3 1 4 1 5 1

Problem J. Subordinates

Time Limit 1000 ms

Mem Limit 524288 kB

Given the structure of a company, your task is to calculate for each employee the number of their subordinates.

Input

The first input line has an integer n : the number of employees. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director of the company.

After this, there are $n - 1$ integers: for each employee $2, 3, \dots, n$ their direct boss in the company.

Output

Print n integers: for each employee $1, 2, \dots, n$ the number of their subordinates.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
5 1 1 2 3	4 1 1 0 0